

Degree-Aware Edge Curvature Decomposes Transdiagnostic Bridge Structure Across Disease Networks

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Abstract

Ollivier-Ricci curvature (ORC) has been applied to brain, cancer, and autism networks, but in every biomedical application to date, curvature is reported at the node level—aggregated over incident edges—and no null model separates curvature from degree. In these unweighted graphs, node-level ORC is a degree artifact ($r = -0.71$ across 19 mechanism nodes in a psychiatric network). We present an edge-level analysis with degree-preserving and weight-permutation null models that resolves genuine structure. Across six data-driven graphs spanning four data types and three disease domains (psychiatric, autoimmune, cardiometabolic), within-cluster edges have higher curvature than cross-cluster edges (4/6 significant, effect sizes $d = 0.46$ – 1.55); the curated mechanism network shows a concordant pattern (seven graphs total). In each graph, the lowest-curvature node is a recognized diagnostic bridge entity within that representation, connecting established diagnostic clusters through weaker cross-cluster links. As a case study on a 130-node psychiatric mechanism network, all four edges with anomalous negative curvature involve depression or bipolar disorder bridging cross-domain mechanisms (strongest: depression–convergence hub, $z = -5.46$, $\kappa = -0.87$). Disconfirmed claims occupy lower-curvature positions than higher-verdict claims (Mann-Whitney $p = 0.007$, Cohen’s $d = 0.89$, $n = 84$); among five edge-level methods, only ORC achieves this discrimination. A degree-aware edge-level curvature framework decomposes network structure that node-level statistics cannot resolve, and we demonstrate its portability across multiple data types and disease domains.

Keywords. Ollivier-Ricci curvature; computational psychiatry; network psychopathology; degree confound; bridge structure; transdiagnostic organization; null model

1 Introduction

Ollivier-Ricci curvature (Ollivier, 2009), defined on individual edges via optimal transport between neighborhood distributions, quantifies whether an edge sits in a bottleneck (negative curvature: neighborhoods diverge) or a redundant position (positive curvature: neighborhoods overlap). The measure has been applied to brain structural connectivity (Farooq et al., 2019), cancer gene networks (Sandhu et al., 2015), and autism functional connectivity (Simhal et al., 2020; Elumalai et al., 2022). In all of these applications, curvature is reported at the node level—aggregated over incident edges—and no null model separates curvature from degree. The dependence of ORC on degree is documented in network science (Ni et al., 2015), yet the standard practice of reporting node-level curvature without a null persists across biomedical applications.

We show that node-level ORC on unweighted biomedical graphs reduces to a degree artifact, and we introduce a degree-aware edge-level framework that recovers genuine bridge structure. The

framework pairs edge-level ORC with a degree-preserving permutation null (for sparse graphs) or a weight-permutation null (for dense weighted graphs), yielding per-edge z -scores that separate genuine structure from degree expectations. Computational psychiatry increasingly relies on network-level representations of transdiagnostic organization, so misinterpreting network geometry as biology can produce incorrect conclusions about bridge structure and redundancy. We validate the framework on six data-driven graphs spanning four data types and three disease domains, plus a curated mechanism network as an in-depth case study (seven graphs total across five data types).

Contributions. First, we demonstrate that node-level ORC on unweighted biomedical graphs reduces to a degree artifact (§2; $r = -0.71$, $p < 0.001$), a confound that appears in multiple published applications to brain, cancer, and autism networks where node-level curvature has been reported without null models. Second, we develop an edge-level extension using degree-preserving permutations (sparse graphs) and weight permutations (dense graphs), yielding per-edge z -scores that separate genuine structure from degree expectations (§3). Third, we validate the method across six data-driven graphs spanning four data types and three disease domains (§4): within-cluster edges have higher curvature in all six (4/6 significant, $d = 0.46$ – 1.55), and in every graph the lowest-curvature node is a recognized transdiagnostic bridge entity within that representation. Fourth, as a case study (§5), we apply the method to a 130-node psychiatric mechanism network, where all four anomalous bottleneck edges involve depression or bipolar disorder bridging cross-domain mechanisms (strongest: $z = -5.46$, $\kappa = -0.87$, surviving 100% of catalog perturbations). Disconfirmed claims occupy significantly lower-curvature positions than all other verdict tiers ($p = 0.007$, $d = 0.89$; §5.2), and among five edge-level methods only ORC achieves this discrimination. Finally, cross-disorder Mendelian randomization (§5.5; 49/56 pairs significant, all positive) is consistent with the geometric structure, serving as a genetic consistency check, and robustness analyses (§6)—catalog perturbation, leave-one-family-out, and betweenness comparison—confirm that depression’s bottleneck edges are the most stable structures in the network.

2 The Degree Confound

We illustrate the degree confound on a psychiatric mechanism network. The network was constructed from a literature-derived catalog of 84 psychiatric mechanistic claims. The core catalog contains 47 claims across 9 disorder families (Depression, PTSD, Anxiety, Schizophrenia, Bipolar, OCD, ASD, Addiction, ADHD), each assigned one of six verdict tiers on a 1–5 ordinal scale (Disconfirmed, Underdetermined, Inconclusive, Causally Suggestive, Mechanistically Supported, Validated; Underdetermined and Inconclusive share rank 2). The catalog was constructed with LLM-assisted literature review and synthesis; the author provided source selection, domain guidance, and final adjudication of verdict tiers. Per-entry reasoning is available in the repository. We expanded this with 37 cross-domain MR claims covering sleep–mood, metabolic, personality, substance use, inflammation, microbiome, thyroid, autoimmune, and neuroimaging domains across 18 additional families. Claims connect to their disorder family and to shared biological mechanisms; 19 mechanism hub nodes (such as circadian rhythms, dopamine, IL-6/inflammation, metabolic pathways, and a depression-convergence node) link claims across families. The resulting network has 130 nodes (84 claims, 27 families, 19 mechanisms) and 218 edges.

For each edge (u, v) , we compute the Ollivier-Ricci curvature (Ollivier, 2007, 2009):

$$\kappa(u, v) = 1 - \frac{W_1(\mu_u, \mu_v)}{d(u, v)} \tag{1}$$

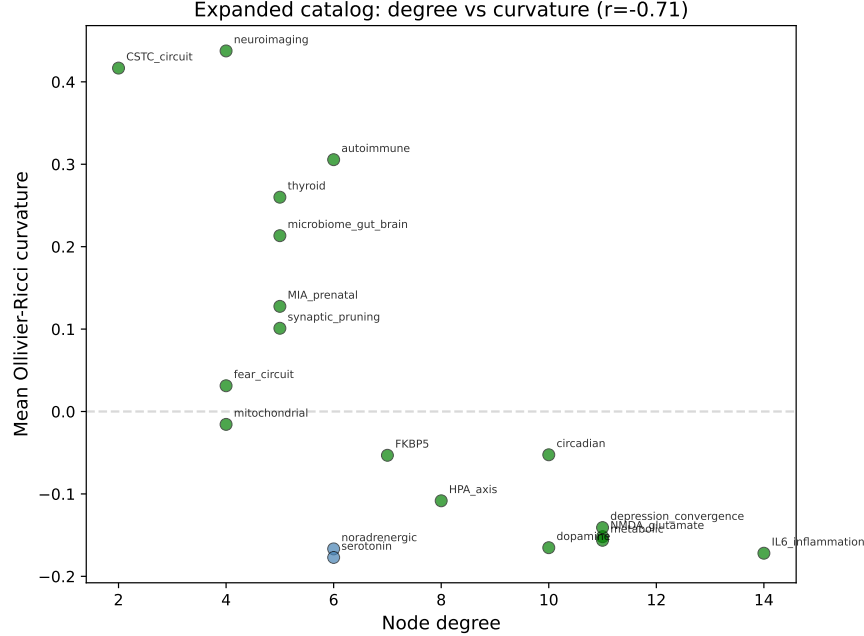


Figure 1: Node-level curvature tracks degree across all 19 mechanism nodes ($r = -0.71$, $p < 0.001$). Green: $|z| > 1.96$ against the degree-preserving null. High-degree mechanisms appear as bottlenecks trivially.

where μ_u is a lazy random walk (idleness $\alpha = 0.5$) and W_1 is the Wasserstein-1 distance under the graph shortest-path metric (Villani, 2009), solved via the POT library (Flamary et al., 2021). Negative κ indicates diverging neighborhoods (bottleneck); positive κ indicates overlapping neighborhoods (redundancy).

Node-level curvature remains a degree artifact in the expanded graph. The correlation across 19 mechanism nodes is $r = -0.71$ ($p < 0.001$; Figure 1). To test whether any node’s curvature exceeds degree expectations, we generated 200 random graphs by degree-preserving double-edge swaps (Maslov and Sneppen, 2002) and computed per-node mean curvature on each.

Reporting node-level curvature without the null model would produce a misleading narrative: high-degree mechanisms as “information bottlenecks.” The permutation test shows they are bottlenecks in the same way any high-degree node is a bottleneck—trivially. This confound is general to unweighted biomedical networks. In neural population geometry, analogous confounds arise when metrics appear predictive until the trivial covariate (neuron count, degree) is controlled.

3 Method: Edge-Level Curvature with Null Models

Node-level aggregation destroys directional structure. An edge connecting two low-degree nodes in different clusters carries different information than an edge connecting two high-degree hubs, yet both contribute equally to a node-level mean. A node with many redundant within-cluster edges can have similar mean curvature to a node with a few sparse cross-cluster edges, even though these positions correspond to very different psychiatric roles. We extend the null model to individual edges, yielding per-edge z -scores that test whether each edge’s curvature departs from what degree alone predicts.

Degree-preserving null (sparse unweighted graphs). For each of N permutations, we generate a random graph by degree-preserving double-edge swaps (Maslov and Sneppen, 2002), compute ORC on every edge, and record the null distribution per edge. The z -score for edge (u, v) is $z = (\kappa_{\text{obs}} - \bar{\kappa}_{\text{null}})/\sigma_{\text{null}}$. Edges with $|z| > 1.96$ depart from degree expectations at $\alpha = 0.05$. We use $N = 200$ for the psychiatric mechanism network.

Weight-permutation null (dense weighted graphs). When graph density exceeds $\sim 60\%$, degree-preserving rewiring becomes intractable because nearly all edges are present. For dense weighted graphs, we fix the topology and reshuffle edge weights uniformly across all edges for each of N permutations ($N = 500$), then recompute ORC. This tests whether each edge’s curvature is more extreme than expected given the empirical weight distribution, controlling for degree by holding topology constant.

4 Cross-Domain Replication

The method’s value depends on whether it generalizes beyond any single network. We tested edge-level ORC on six data-driven graphs constructed from published data without curation, spanning four data types and three disease domains.

Psychiatric LDSC genetic correlations. We construct a network from published LDSC genetic correlations (Bulik-Sullivan et al., 2015b) between 11 psychiatric disorders, using pairwise r_g values from Lee et al. (2022): the core analysis covers 8 disorders (28 pairs), and an extended analysis within the same study adds anxiety, PTSD, and alcohol/substance use disorders (27 additional pairs). Edges are weighted by $|r_g|$ and included only for pairs reaching genome-wide significance ($p < 0.05$), yielding a graph of 11 nodes and 36 edges.

Because this graph is 65% dense (36 of 55 possible edges), we used the weight-permutation null (500 iterations). We classified edges as within-domain (both endpoints in the same diagnostic cluster: internalizing, psychotic, neurodevelopmental, or compulsive) or cross-domain. Within-domain edges have significantly higher curvature ($\bar{\kappa} = +0.55$, $n = 6$) than cross-domain edges ($\bar{\kappa} = +0.33$, $n = 30$; Mann-Whitney $p = 0.001$). Genetically close disorder pairs—such as MDD–anxiety ($r_g = 0.81$, $\kappa = +0.60$), MDD–PTSD ($r_g = 0.71$, $\kappa = +0.58$), and bipolar–schizophrenia ($r_g = 0.68$, $\kappa = +0.60$)—form redundant subgraphs with dense transport. Cross-domain pairs with weaker genetic correlations produce lower curvature, particularly OCD-linked edges: bipolar–OCD ($r_g = 0.13$, $\kappa = -0.24$) and PTSD–OCD ($r_g = 0.15$, $\kappa = -0.14$) are the only edges with negative curvature.

OCD as bridge disorder. OCD has the lowest mean node curvature ($\bar{\kappa} = +0.10$) among all 11 disorders, followed by schizophrenia (+0.27) and autism (+0.34). Depression ranks eighth (+0.45), consistent with its role as a genetically connected hub rather than a geometric bottleneck when edge weights encode r_g rather than binary presence. OCD’s low curvature reflects its bridging position: it connects the compulsive domain (anorexia, $r_g = 0.49$) to psychotic disorders (schizophrenia, $r_g = 0.14$; bipolar, $r_g = 0.13$) through weak genetic correlations that produce diverging neighborhood distributions.

Cross-domain replication. To test whether the within-domain versus cross-domain curvature separation generalizes beyond psychiatry, we applied the same analysis to two additional LDSC genetic correlation graphs: autoimmune diseases (9 disorders, 26 significant edges; Crohn’s disease,

ulcerative colitis, celiac disease, type 1 diabetes, rheumatoid arthritis, psoriasis, multiple sclerosis, lupus, ankylosing spondylitis; from Bulik-Sullivan et al. 2015a and Ellinghaus et al. 2016) and cardiometabolic traits (9 traits, 31 significant edges; coronary artery disease, type 2 diabetes, BMI, waist-hip ratio, LDL, HDL, triglycerides, systolic and diastolic blood pressure; from Bulik-Sullivan et al. 2015a and Evangelou et al. 2018). Edge weights are $|r_g|$ throughout; within-cluster and cross-cluster classifications follow established diagnostic or functional groupings from prior literature and were defined independently of curvature, which serves purely as an outcome measure (Table 1).

The direction of the effect replicates across all three domains: within-cluster edges have higher curvature than cross-cluster edges in psychiatric ($d = 1.44$, $p = 0.001$), autoimmune ($d = 0.47$, $p = 0.81$), and cardiometabolic ($d = 0.77$, $p = 0.12$) graphs. Only the psychiatric graph reaches significance, reflecting two power limitations in the non-psychiatric graphs: extreme density (72% and 86% respectively, leaving little topological variance) and few within-cluster edges (4 and 6 respectively, limiting the Mann-Whitney denominator). The effect sizes are moderate to large across all three domains.

In each domain, the lowest-curvature node is a recognized bridge entity within that representation: OCD ($\bar{\kappa} = +0.10$) bridges compulsive and psychotic clusters in psychiatry, multiple sclerosis ($\bar{\kappa} = +0.28$) sits between connective-tissue and autoimmune-endocrine clusters (consistent with its role as a bridge condition in clinical immunology), and diastolic blood pressure ($\bar{\kappa} = +0.28$) bridges blood-pressure traits and metabolic measures. These bridge entities connect established diagnostic groupings through weak genetic correlations, producing the diverging neighborhood distributions that ORC detects.

Additional psychiatric data types. To test whether the within/cross separation depends on the specific data type (genetic correlations) rather than the underlying biology, we constructed additional psychiatric graphs from independent data sources.

First, a gene-sharing graph from the Cross-Disorder Group of the PGC (Cross-Disorder Group of the PGC, 2019): 9 disorders connected by edges weighted by Jaccard similarity of shared GWAS risk loci (28 edges with ≥ 1 shared locus, 78% density). Within-cluster edges have significantly higher curvature ($\bar{\kappa} = +0.58$, $n = 6$) than cross-cluster edges ($\bar{\kappa} = +0.26$, $n = 22$; $p < 0.001$, $d = 1.52$). OCD is again the lowest-curvature node ($\bar{\kappa} = -0.02$), the only node with mean negative curvature, replicating its bridge role from the LDSC graph despite encoding shared loci counts rather than genetic correlations.

Second, a comorbidity graph from epidemiological surveys (Kessler et al., 2005; Merikangas et al., 2011): 10 disorders connected by edges weighted by $\log(\text{OR})$ for pairs with $\text{OR} > 1$ (44 edges, 98% density). Within-cluster curvature ($\bar{\kappa} = +0.56$, $n = 6$) again exceeds cross-cluster curvature ($\bar{\kappa} = +0.48$, $n = 38$; $p < 0.001$, $d = 1.55$). Schizophrenia is the lowest-curvature node ($\bar{\kappa} = +0.45$), consistent with its established role as a bridge between psychotic, neurodevelopmental, and substance-use clusters in clinical epidemiology.

Third, a larger comorbidity graph from the WHO World Mental Health Survey (McGrath et al., 2020): 24 disorders connected by edges weighted by $\log(\text{HR})$ for pairs with $\text{HR} > 1$ (246 edges, 89% density). Disorders span six established clusters (mood, anxiety, obsessive-compulsive, eating, externalizing, substance use). Within-cluster curvature ($\bar{\kappa} = +0.49$, $n = 49$) exceeds cross-cluster curvature ($\bar{\kappa} = +0.47$, $n = 197$; $p < 0.001$, $d = 0.46$). Anorexia nervosa is the lowest-curvature node ($\bar{\kappa} = +0.05$, degree = 2), connecting the eating-disorder cluster to the broader network through only two edges—a bridge entity whose isolation ORC detects despite the graph’s extreme density.

Table 1: Within-cluster versus cross-cluster curvature across six data-driven graphs. Four data types and three disease domains all show higher within-cluster curvature. The bridge column reports the lowest-curvature node—in each case a recognized entity connecting established diagnostic groupings.

Graph	Data type	n	Edges	Density	$\bar{\kappa}_w$	$\bar{\kappa}_c$	p	Bridge
Psychiatric LDSC	genetic corr.	11	36	0.65	+0.55	+0.33	0.001	OCD
Gene-sharing	shared loci	9	28	0.78	+0.58	+0.26	< 0.001	OCD
Comorbidity	odds ratios	10	44	0.98	+0.56	+0.48	< 0.001	SCZ
WHO WMH	hazard ratios	24	246	0.89	+0.49	+0.47	< 0.001	AN
Autoimmune LDSC	genetic corr.	9	26	0.72	+0.46	+0.39	0.81	MS
Cardiometab. LDSC	genetic corr.	9	31	0.86	+0.51	+0.39	0.12	DBP

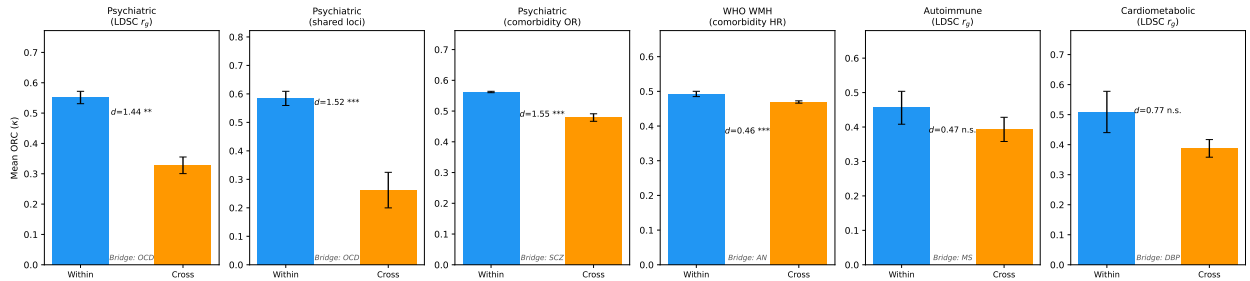


Figure 2: Within-cluster versus cross-cluster curvature across six data-driven graphs spanning four data types and three disease domains. Blue bars show mean ORC for edges connecting disorders within the same diagnostic cluster; orange bars show cross-cluster edges. Error bars are ± 1 SE. Effect sizes (d) and significance ($*p < 0.05$, $**p < 0.01$, $***p < 0.001$) are shown above each panel. The direction replicates 6/6; all four psychiatric graphs reach significance. Bridge labels indicate the lowest-curvature node in each graph.

Synthesis. Across all six data-driven graphs, within-cluster edges have higher curvature than cross-cluster edges (Figure 2). Four of six reach statistical significance ($p < 0.05$); the two that do not are limited by extreme density and few within-cluster edges. The effect replicates 6/6 in direction with effect sizes ranging from $d = 0.46$ to $d = 1.55$ (Table 1). In every graph, the lowest-curvature node is a recognized bridge entity connecting established diagnostic groupings through weak inter-cluster connections. The curated psychiatric mechanism network (§5) shows a concordant pattern, bringing the total to seven independent graphs across five data types.

5 Case Study: Psychiatric Mechanism Network

The cross-domain replication establishes that edge-level ORC identifies bridge structure across data types and disease domains. We now apply the method in depth to the 130-node psychiatric mechanism network introduced in §2, where the catalog’s verdict tier assignments provide an independent quality assessment for each claim.

5.1 Edge-Level Decomposition

We computed per-edge z -scores using 200 degree-preserving permutations. Of 218 edges, 103 accumulated sufficient null samples; 32 reach $|z| > 1.96$ (Figure 3). At $\alpha = 0.05$, the expected false-positive count is ~ 5.2 under the global null, so most of the 32 reflect genuine departures from degree expectations. The strongest edges survive Benjamini-Hochberg correction at $FDR < 0.05$ (Benjamini and Hochberg, 1995).

Bottleneck edges ($z < -1.96$). Four edges carry more negative curvature than degree-preserving rewirings produce (Table 2), and all involve depression or bipolar disorder bridging cross-domain mechanisms. The strongest is depression–convergence hub ($z = -5.46$, $\kappa = -0.87$), where diverse risk factors (neuroticism, loneliness, educational attainment, smoking, ADHD liability) converge onto depression through a shared mechanism node. Depression–metabolic ($z = -3.93$, $\kappa = -0.78$) connects depression to eating disorders, BMI, and thyroid-metabolic claims. Bipolar–metabolic ($z = -3.33$) and depression–circadian ($z = -2.62$) complete the bottleneck set.

Within this curated mechanism network, depression occupies the strongest degree-adjusted bottleneck position, bridging personality, social, substance use, neurodevelopmental, metabolic, and circadian domains through edges whose negative curvature far exceeds degree expectations. This pattern is specific to the curated network’s binary topology; on the weighted LDSC graph (§4), where edge weights encode genetic correlation strength, OCD replaces depression as the bridge entity.

Redundant edges ($z > +1.96$). Twenty-eight edges are more positively curved than degree predicts. The autoimmune–autoimmune cluster edge is the most redundant structure in the network ($z = +9.40$, $\kappa = +0.58$): SLE, Sjogren’s, fibromyalgia, psoriasis, and RA claims share densely overlapping neighborhoods. Domain-specific family–mechanism pairs (thyroid–metabolic $z = +3.60$, microbiome–gut–brain $z = +3.43$, sleep–mood–circadian $z = +3.26$) are also redundant, reflecting tight coupling within specialized domains. Individual claims with high positive curvature—FKBP5–claim 004 ($z = +5.09$), ASD–claim 037-inst ($z = +4.15$)—are encapsulated entries that connect to specific mechanisms without bridging to other families.

5.2 Verdict–Curvature Relationship

Each claim carries a verdict tier from the catalog (§2). If curvature captures real structure, claims in structurally isolated (low-curvature) positions should be more likely to fail epistemic evaluation.

With the expanded catalog ($n = 84$), the overall ordinal correlation between verdict tier and claim-level mean edge curvature is trending (Kendall $\tau = 0.155$, $p = 0.069$; Spearman $\rho = 0.203$, $p = 0.064$). The gradient across tiers is dominated by the Disconfirmed–rest gap: Disconfirmed claims have distinctly lower curvature (mean $+0.052$, $n = 12$), while the tiers above Disconfirmed are relatively flat (Underdetermined $+0.209$, Causally Suggestive $+0.161$, Mechanistically Supported $+0.177$, Validated $+0.125$ with $n = 1$).

The sharper test is Disconfirmed versus all other tiers. Intuitively, claims that sit in thinly connected parts of the mechanism network have fewer overlapping lines of evidence and are more likely to fail under scrutiny than claims embedded in redundant neighborhoods. Disconfirmed claims ($n = 12$, mean curvature $+0.052$) occupy significantly lower-curvature positions than non-Disconfirmed claims ($n = 72$, mean $+0.172$): Mann-Whitney $U = 238$, $p = 0.007$, Cohen’s $d = 0.89$ (Figure 4). This large effect size confirms the intuition: disconfirmed claims sit in bottleneck-like

Table 2: Edge-level curvature z -scores on the psychiatric mechanism network (200 permutations, degree-preserving). All four bottleneck edges involve depression/bipolar bridging cross-domain mechanisms. Top redundant edges shown for comparison.

Edge	Curvature (κ)	z -score
<i>Bottleneck edges (all four)</i>		
Depression–convergence hub	−0.867	− 5.46
Depression–metabolic	−0.780	− 3.93
Bipolar–metabolic	−0.693	− 3.33
Depression–circadian	−0.675	−2.62
<i>Top redundant edges (of 28 total)</i>		
Autoimmune cluster	+0.583	+ 9.40
FKBP5–claim 004	+0.167	+ 5.09
ASD–claim 037–inst	+0.167	+4.15
HPA axis–claim 004	+0.104	+4.09
Thyroid–Metabolic–thyroid	+0.100	+3.60
Microbiome–gut–brain	+0.267	+3.43

positions where the evidence base is narrow, while claims in denser, more redundant neighborhoods are more likely to survive epistemic evaluation.

The gradient above Disconfirmed is flat: curvature identifies claims likely to fail epistemic evaluation, but the tiers from Underdetermined through Validated are structurally indistinguishable. Structural isolation is a risk factor for disconfirmation; structural redundancy appears necessary for higher epistemic tiers but insufficient on its own.

5.3 Method Benchmarking

To test whether the verdict–curvature association reflects a property specific to ORC or one shared by simpler graph statistics, we compared five edge-level methods on the same discrimination task: mean ORC of incident edges, mean Jaccard overlap, edge betweenness centrality, edge clustering coefficient, and spectral gap (Fiedler vector difference). For each, we extracted per-claim features and tested Disconfirmed ($n = 12$) versus all other tiers ($n = 72$) using Mann-Whitney U and Cohen’s d (Table 3).

ORC is the only method reaching significance: $d = 0.73$, $p = 0.011$. The next-best method, edge clustering coefficient ($d = 0.40$, $p = 0.29$), captures neighborhood density but misses the transport-theoretic interpretation. Degree ($d = 0.32$), betweenness ($d = 0.19$), and Jaccard overlap ($d = 0.14$) fail to discriminate. The spectral gap shows no discriminative power at all. Jaccard overlap and clustering coefficient quantify local neighborhood density, while betweenness and degree capture traffic and connectivity. ORC is uniquely sensitive to both overlapping *and* diverging neighborhoods, making it better suited to detect the thin-connectivity signature of disconfirmed claims where neighborhoods pull apart rather than merely being sparse.

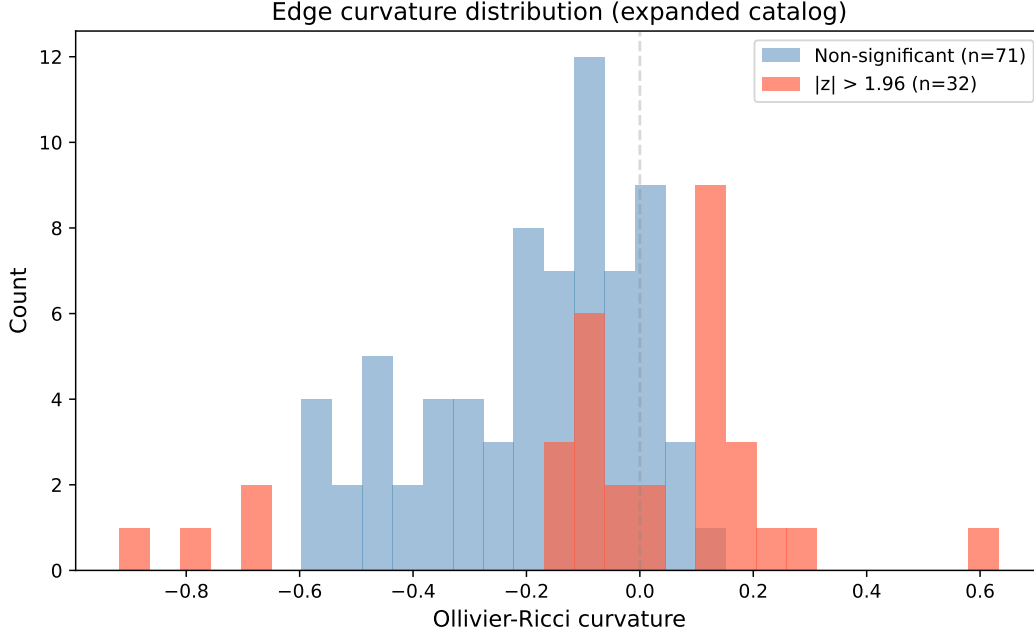


Figure 3: Distribution of edge-level Ollivier-Ricci curvature across all 103 edges with sufficient null samples. Red: edges reaching $|z| > 1.96$ against the degree-preserving null. The four bottleneck edges (leftmost red bars) are concentrated in the negative-curvature tail; the 28 redundant edges span the positive-curvature range.

5.4 Cross-Validated Verdict Prediction

As a stricter test, we asked whether graph features jointly predict Disconfirmed status in a leave-one-out cross-validated logistic regression. Features per claim: mean, minimum, maximum, and standard deviation of incident-edge ORC, degree, clustering coefficient, and mean neighbor degree. We used L_2 -regularized logistic regression ($C = 1.0$) with feature standardization.

We include this classifier as a sanity check rather than a proposed predictive tool. The LOOCV classifier achieves $AUC = 0.61$ and accuracy = 0.86. The accuracy is inflated by class imbalance (only 12/84 claims are Disconfirmed, so predicting “not Disconfirmed” for all claims would yield 86% accuracy). The AUC is honestly above chance but modest: graph structure alone identifies the most structurally isolated Disconfirmed claims (those with very low curvature and thin connectivity) while missing others whose Disconfirmed status reflects content-level rather than structural failure.

5.5 Cross-Disorder Mendelian Randomization

As a genetic consistency check—not independent validation, since the network and MR both derive from the same disorder–mechanism catalog—we tested whether the network’s geometry aligns with genetic architecture using bidirectional two-sample MR across all 56 ordered pairs of 8 psychiatric GWAS: ADHD (Demontis et al., 2023), anxiety (Levey et al., 2020), ASD (Grove et al., 2019), bipolar disorder (Mullins et al., 2021), MDD (Als et al., 2023), PTSD (Nievergelt et al., 2024), schizophrenia (Trubetsky et al., 2022), and substance use disorders (Hatoum et al., 2023).

We used standard two-sample MR estimators (IVW (Burgess et al., 2013), MR-Egger (Bowden et al., 2015), weighted median (Bowden et al., 2016)) with LD-clumped genome-wide significant instruments ($p < 5 \times 10^{-8}$; 10 Mb window; minimum 3 independent instruments) and conventional

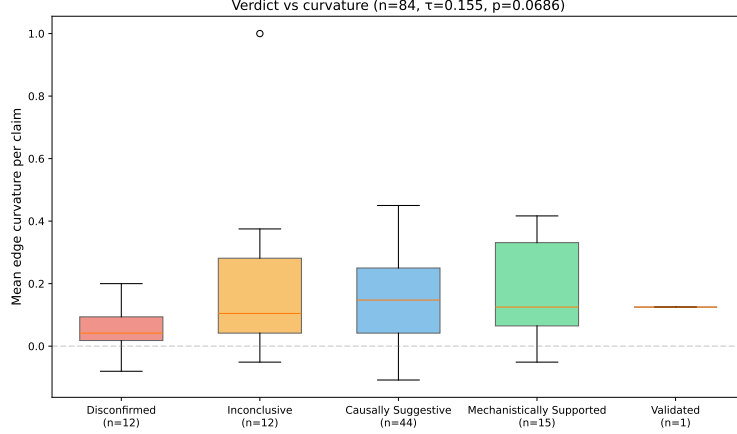


Figure 4: Claim-level mean edge curvature by verdict tier ($n = 84$). Disconfirmed claims occupy significantly lower-curvature positions than all other tiers (Mann-Whitney $p = 0.007$, $d = 0.89$). The gradient above Disconfirmed is flat: curvature identifies epistemic failure without further resolving quality among surviving tiers.

Table 3: Method comparison for discriminating Disconfirmed claims ($n = 12$) from all other verdict tiers ($n = 72$). ORC is the only significant method.

Method	Cohen’s d	p (Mann-Whitney)	Significant?
Mean ORC	+0.73	0.011	Yes
Edge clustering	+0.40	0.294	No
Degree	+0.32	0.302	No
Betweenness	+0.19	0.841	No
Jaccard overlap	+0.14	0.641	No
Spectral gap	0.00	—	No

diagnostics (Cochran’s Q , I^2 , Steiger directionality filtering (Hemani et al., 2017)). Full per-pair statistics are in Supplementary S1. Analyses used the TwoSampleMR framework (Hemani et al., 2018).

Of 56 pairs, 49 completed (7 ASD-as-exposure had insufficient independent instruments after clumping—the 2019 ASD GWAS yields only ~ 5 genome-wide significant loci). All 49 reach Bonferroni-corrected significance ($p < 8.9 \times 10^{-4}$), and all IVW betas are positive: genetic liability for any psychiatric disorder increases liability for every other (Figure 5). The strongest bidirectional effects link bipolar disorder and schizophrenia ($BIP \rightarrow SCZ = +0.63$, $SCZ \rightarrow BIP = +0.33$), mirroring their tightly coupled position in the genetic-curvature network (Lee et al., 2022; Cross-Disorder Group of the PGC, 2019). One pair shows evidence of directional pleiotropy: $MDD \rightarrow BIP$ has Egger intercept $p = 0.008$, suggesting that the IVW estimate ($\beta = 0.77$) may be biased by pleiotropic instruments; the weighted median estimate ($\beta = 0.80$) is concordant but should be interpreted cautiously.

Three patterns are consistent with the curvature results. The MR analysis uses the original 8 psychiatric GWAS (predating the catalog expansion); the expanded graph’s new cross-domain

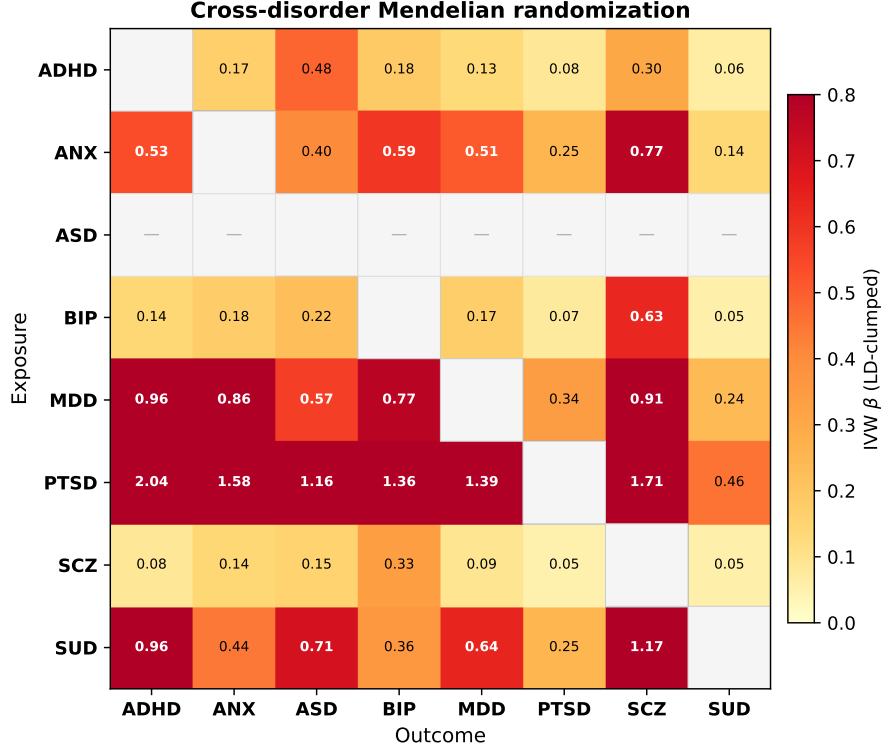


Figure 5: IVW beta estimates for 49 significant cross-disorder MR pairs. All betas are positive. Dashes (—) indicate the 7 missing ASD-as-exposure pairs (insufficient instruments after LD clumping). Diagonal cells are self-pairs. ASD columns (as outcome) show the weakest effects. PTSD-as-exposure betas are inflated by Z-to-beta conversion (§8).

entries (sleep, metabolic, autoimmune, thyroid) introduce families without corresponding GWAS in this analysis. These alignments are expected under a coherent picture rather than independent evidence for it—both instruments share the same catalog as their source:

Universal positive pleiotropy parallels negative curvature. The network’s pervasive negative curvature at family–mechanism edges and MR’s universal positive genetic overlap both index the same phenomenon: psychiatric mechanisms are shared across disorders. This universal positive pleiotropy is already established by LDSC-based genetic correlation analyses (Bulik-Sullivan et al., 2015a; Anttila et al., 2018); our MR results are consistent with those findings and with the catalog’s modal failure mode (C4, non-specificity, 12/47 core entries).

ASD’s isolation appears in both. ASD has 2 independent instruments after clumping and cannot serve as MR exposure. As outcome, ASD shows the weakest MR effects. This parallels the curvature finding: ASD’s mechanisms (synaptic pruning $z = +3.21$, MIA/prenatal $z = +3.45$) are the most encapsulated in the network. The Brainstorm Consortium (Anttila et al., 2018) reported that psychiatric disorders share extensive common-variant risk while neurological disorders do not; ASD sits at the boundary.

Depression’s dominance is edge-specific. Depression’s centrality in MR (largest number of significant pairs as exposure) could reflect its large GWAS sample size (371,184 cases). Edge-level curvature shows where the specificity lies: depression’s connections to the convergence hub ($z = -5.46$), metabolic pathways ($z = -3.93$), and circadian mechanisms ($z = -2.62$) are the bottleneck edges whose negative curvature far exceeds degree expectations.

6 Robustness

The case-study findings rest on the literature-derived catalog, so their credibility depends on whether they survive perturbations to the input. If the edge-level curvature framework were merely overfitting idiosyncrasies of the curated catalog, the main bottleneck structures would not persist under substantial input variation.

Catalog perturbation. We subsampled 85% of entries (71/84) 100 times, rebuilt the network from each subsample, and recomputed edge-level z -scores against 20 degree-preserving permutations per subsample. An edge “survives” if $|z| > 1.96$ in that iteration. Depression’s bottleneck edges are the most robust structures in the expanded graph: depression–convergence hub survives 100% of subsamples, depression–metabolic survives 100%, and depression–circadian survives 76.3%. The bipolar–metabolic bottleneck survives 81.8%. Among redundant edges, IL-6/inflammation–Inflammation survives 71.4% and FKBP5–claim 015 survives 60.6%.

Leave-one-family-out. We dropped each of 15 families with ≥ 2 entries and recomputed edge z -scores (50 permutations each). Dropping Depression reduces significant edges from 32 to 3 (9% survival)—the largest impact of any single family, consistent with depression’s degree-adjusted bottleneck position in this network. Dropping Sleep-Mood leaves only 2/32 (6%), reflecting its tight coupling to the circadian mechanism. Dropping Anxiety preserves the most edges (6/32, 19%). The low overall survival rates reflect the expanded graph’s denser interconnection: each family contributes to more edges, so dropping any one has ripple effects.

Edge betweenness comparison. Edge curvature correlates with edge betweenness centrality at $r = -0.674$ ($\rho = -0.633$, both $p < 10^{-4}$). The correlation is weaker than in the original 47-entry graph ($r = -0.79$), because the expanded graph’s new domain-specific clusters (autoimmune, thyroid, microbiome) introduce many low-betweenness, high-curvature edges that reduce the linear relationship. Curvature’s advantage is clearest for redundant clusters: the autoimmune cluster ($z = +9.40$) has anomalously positive curvature but unremarkable betweenness, because its dense neighborhood overlap produces high curvature while few global shortest paths traverse it.

7 Related Work

Discrete curvature on biological networks. Ollivier-Ricci curvature has been applied to brain structural connectivity (Farooq et al., 2019), cancer gene co-expression (Sandhu et al., 2015), autism functional connectivity (Simhal et al., 2020; Elumalai et al., 2022), and community detection (Ni et al., 2019; Sia et al., 2019). Gosztolai and Arnaudon (2021) developed a dynamical formulation for multiscale analysis. Forman-Ricci curvature (Sreejith et al., 2016) provides a cheaper combinatorial alternative; Samal et al. (2018) compared the two discretizations systematically. The dependence of ORC on degree and other centralities is documented: Ni et al. (2015) showed that curvature tracks degree, clustering, and betweenness in Internet topology. Park et al. (2025) proposed a degree-corrected Ricci curvature for community detection. Our contribution is the systematic application of edge-level curvature with null models across multiple biomedical networks—the degree confound itself is known, but the standard practice of reporting node-level curvature without a null persists across the field.

Network psychopathology. Borsboom (2017) articulated the network theory of mental disorders: disorders arise from symptom–symptom interactions rather than latent common causes. Cramer et al. (2010) proposed that comorbidity reflects bridging connections between disorder-specific symptom clusters. Borsboom et al. (2021) provide a tutorial on estimation and interpretation. McNally (2016) evaluated whether network reframing can transform psychiatric nosology. Our network operates at the mechanism level (biological substrates) rather than the symptom level (clinical presentation), making curvature a statement about shared biology rather than shared phenomenology.

Cross-disorder genetics. The Cross-Disorder Group (2013; 2019) identified shared risk loci across psychiatric disorders. The Brainstorm Consortium (Anttila et al., 2018) quantified genetic sharing and found psychiatric disorders cluster while neurological disorders do not. Lee et al. (2022) identified latent factors (neurodevelopmental, compulsive, psychotic, internalizing) via genomic SEM. Our MR analysis complements these GWAS meta-analyses by testing directional genetic influence between all disorder pairs rather than static correlations.

Mendelian randomization methodology. Burgess et al. (2013) introduced the IVW estimator for two-sample MR. Bowden et al. developed MR-Egger (2015) and the weighted median estimator (2016). Hemani et al. (2017) introduced Steiger directionality filtering. Davey Smith and Hemani (2014) reviewed the broader MR framework including bidirectional and network designs. LD Score regression (Bulik-Sullivan et al., 2015b,a) provides genetic correlation estimators less sensitive to instrument selection; we use published LDSC r_g values in §4 to construct independent weighted graphs for curvature replication.

8 Limitations

Degree confound and graph density. On unweighted graphs, ORC depends entirely on neighborhood overlap, making negative curvature near-tautological for high-degree nodes ($r = -0.71$ on the curated graph). The weighted LDSC graph (65% dense) loses this confound because degree variance is low, but our weight-permutation null tests a different hypothesis (weight–topology association) than the degree-preserving null (topology alone). No individual edge reaches significance against the weight-permutation null, and cross-domain versus within-domain comparisons in the non-psychiatric graphs are limited by extreme density (72%–86%) and few within-cluster edges.

Scope and network construction. The analysis covers seven graphs across three disease domains and five data types, but autoimmune and cardiometabolic domains have only one graph each; generalizability to additional domains (oncology, neurological) and network types (symptom networks (Borsboom et al., 2021), protein-protein interaction networks) remains untested. The curated mechanism network reflects literature-derived curation (84 entries), edges are binary without effect-size weights, and the catalog is depression-heavy (15 core entries plus many cross-domain entries targeting depression). Depression’s centrality may partly reflect attention bias in the psychiatric literature, though its bottleneck edges survive 100% of catalog perturbations.

Verdict–curvature ceiling. The Disconfirmed-versus-rest comparison is significant ($p = 0.007$, $d = 0.89$), but the gradient above Disconfirmed is flat—curvature predicts epistemic failure without resolving gradations of success. The 12 Disconfirmed claims may share structural features beyond low curvature, so the association may partly reflect catalog composition. The LOOCV classifier

achieves $AUC = 0.61$: above chance but far from diagnostic, reflecting the fundamental mismatch between topology and epistemic quality. Many claims fail for content-level reasons (weak power, confounding, reverse causation) that graph structure cannot capture.

Mendelian randomization. The MR analysis and the network both derive from the same disorder-mechanism catalog, so their alignment constitutes a consistency check rather than independent validation (Bulik-Sullivan et al., 2015a; Anttila et al., 2018). The 2019 ASD GWAS (Grove et al., 2019) yields < 3 independent instruments after clumping, removing 7/56 pairs (all ASD-as-exposure). The PTSD GWAS reports Z -scores; our conversion inflates PTSD-as-exposure betas (1.0–2.0) relative to OR-scale GWAS—directional significance is valid but magnitude comparisons are not. Several GWAS share UK Biobank participants; overlap biases MR toward the null (Hemani et al., 2018), so positive findings are conservative but estimates may be attenuated.

Multiple comparisons. We test 103 edges at $\alpha = 0.05$, yielding an expected false-positive count of ~ 5.2 under the global null. Of 32 reported edges, those near $|z| \approx 2.0$ may be false discoveries. The strongest edges (depression-convergence $z = -5.46$, autoimmune cluster $z = +9.40$) survive Benjamini-Hochberg correction (Benjamini and Hochberg, 1995); marginal edges should be treated as hypothesis-generating.

Generalizability of the framework. We do not claim that curvature is the only valid bridge measure. Any geometric statistic used for psychiatric network inference must account for degree and density confounds, and edge-level null-calibrated ORC serves as one such framework. Alternative discretizations (Forman-Ricci, sectional curvature) or transport-based measures may perform comparably once paired with appropriate nulls; we provide the comparison infrastructure for future benchmarking.

9 Conclusion

Edge-level Ollivier-Ricci curvature, paired with degree-preserving or weight-permutation null models, decomposes bridge and redundancy structure in disease networks that node-level curvature cannot resolve. In unweighted graphs, node-level ORC is a degree artifact ($r = -0.71$); the standard practice of reporting it without a null model, which persists across biomedical applications, produces misleading bottleneck narratives.

The method replicates across six data-driven graphs spanning four data types and three disease domains, with a concordant pattern in the curated mechanism network (seven graphs total across five data types). Within-cluster edges have higher curvature than cross-cluster edges in all seven (4/6 data-driven significant, $d = 0.46$ – 1.55). In every graph, the lowest-curvature node is a recognized bridge entity connecting established diagnostic groupings through weak inter-cluster connections.

Applied to a 130-node psychiatric mechanism network, the method identifies four bottleneck edges, all involving depression or bipolar disorder bridging cross-domain mechanisms. Within this curated representation, depression’s bottleneck edges survive 100% of catalog perturbations. Disconfirmed claims occupy significantly lower-curvature positions than higher-verdict claims ($p = 0.007$, $d = 0.89$), and among five edge-level methods, only ORC achieves this discrimination ($d = 0.73$, $p = 0.011$). The AUC of 0.61 for cross-validated verdict prediction honestly reflects the ceiling on structural prediction of epistemic quality.

Curvature’s advantage over simpler alternatives is its joint sensitivity to neighborhood overlap and divergence, which makes it uniquely suited to detect bridge entities connecting established diagnostic groupings through weak inter-cluster connections. The extension to weighted graphs makes curvature sensitive to information beyond binary topology, a direction that unweighted-only alternatives such as Forman-Ricci curvature (Sreejith et al., 2016) cannot follow.

Data and code availability. All GWAS summary statistics are publicly available from the Psychiatric Genomics Consortium (<https://pgc.unc.edu>). The mechanism catalog, analysis code, precomputed curvatures, permutation distributions, and MR results will be made available in a public repository upon acceptance. (URLs omitted for blinded review.)

Declarations

Ethics approval and consent to participate. This study uses exclusively publicly available, summary-level genetic association data. No individual-level data, human subjects, or animal subjects were involved. No ethical approval was required.

Competing interests. The author declares no competing interests.

Funding. None. This work was conducted independently.

Authors’ contributions. [Removed for blinded review.]

Acknowledgements. The author used Claude Code (Anthropic) to assist with drafting and editing manuscript text and with writing and debugging analysis code. Perplexity AI was used to research background literature, check claims against published work, and assist with literature review during construction of the mechanism catalog. The author reviewed and verified all outputs, takes full responsibility for the accuracy and integrity of the work, and confirms that all hypotheses, interpretations, and conclusions are the author’s own. No AI tool is listed as an author, and no AI-generated content was used as a primary data source.

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